

Carbon Capture for Freight Rail

Life Cycle Assessment of Carbon Capture Railcars Integrated into Existing Freight Rail Infrastructure

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Agenda

- 01 Background: U.S. Freight Rail
- 02 Freight Rail Emission Characteristics
- 03 Rail-Based Carbon Capture
- 04 Technology Selection
- 05 Life Cycle Inventory & Assessment
- 06 Conclusions

Background: U.S. Freight Rail Economic Impact

1.7 Billion

Tons of Goods

Moved annually across the U.S. freight rail network

140,000

Miles of Rail

Spanning the nation's freight rail infrastructure

\$233.4 Billion

Total Economic Output

Generated by the U.S. freight rail industry

150,000+

Direct Employees

Workers employed across the freight rail sector

Background: U.S. Freight Rail Environmental Impact

1.7B

Tons of Goods Moved
Annually

140K

Miles of Rail Network

35M

Metric Tons CO₂ per Year

Low Carbon Transportation

- Among the least carbon-intensive freight modes, on a ton-mile basis, rivaled only by maritime shipping

Emissions Challenge

- Freight rail produces ~35 million metric tons of CO₂ annually and can be considered “hard-to-abate” due to the nature of the US freight system.
- Locomotives have long lifetimes, and as a result, older locomotives are only subject to older emission standards. Older locomotives emit high levels of SO_x, NO_x, and particulate matter

Decarbonization Barriers

- Electrification is prohibitively expensive
- Policy, private ownership, and funding constraints limit large-scale intervention

Background: U.S. Freight Rail Environmental Impact

EPA emission standards: Tier 0 (2000) vs. Tier 4 (2015) — Standard in g/bhp-hr

Line Haul Locomotives



Switch Locomotives

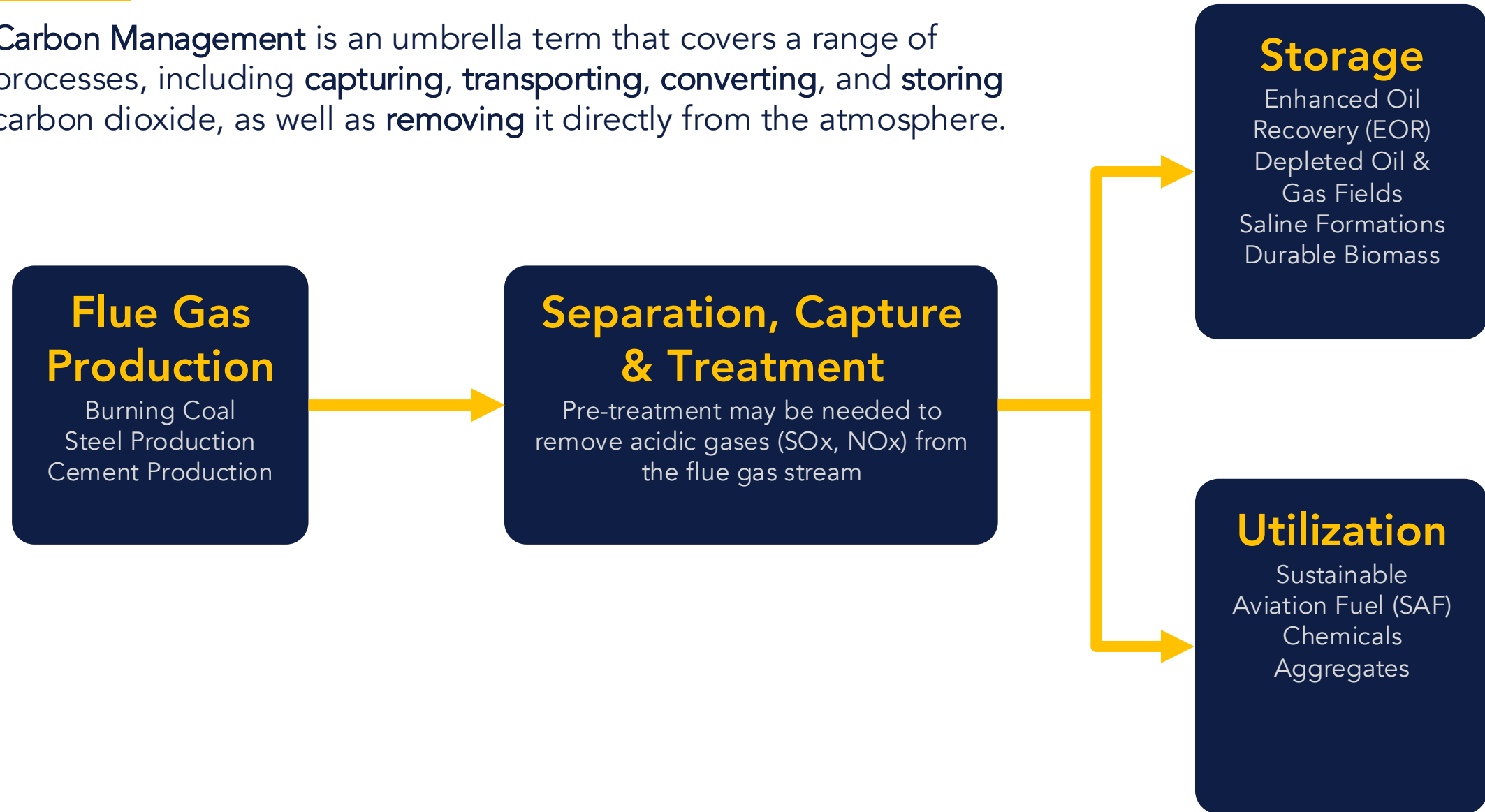


Percent of Work by Locomotive Emission Tier (South Coast Air Basin 2014)



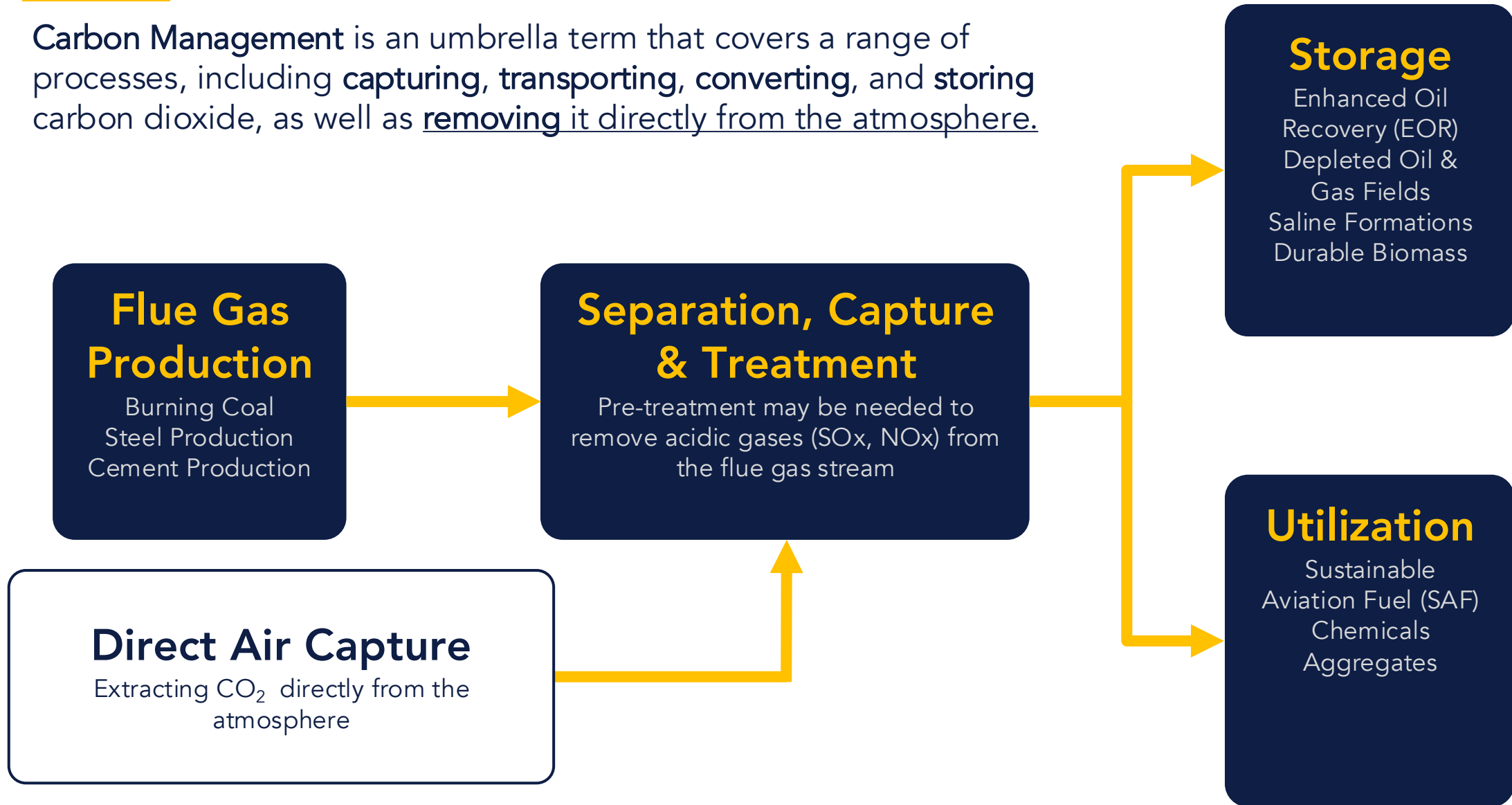
Carbon Management Overview

Carbon Management is an umbrella term that covers a range of processes, including **capturing**, **transporting**, **converting**, and **storing** carbon dioxide, as well as **removing** it directly from the atmosphere.

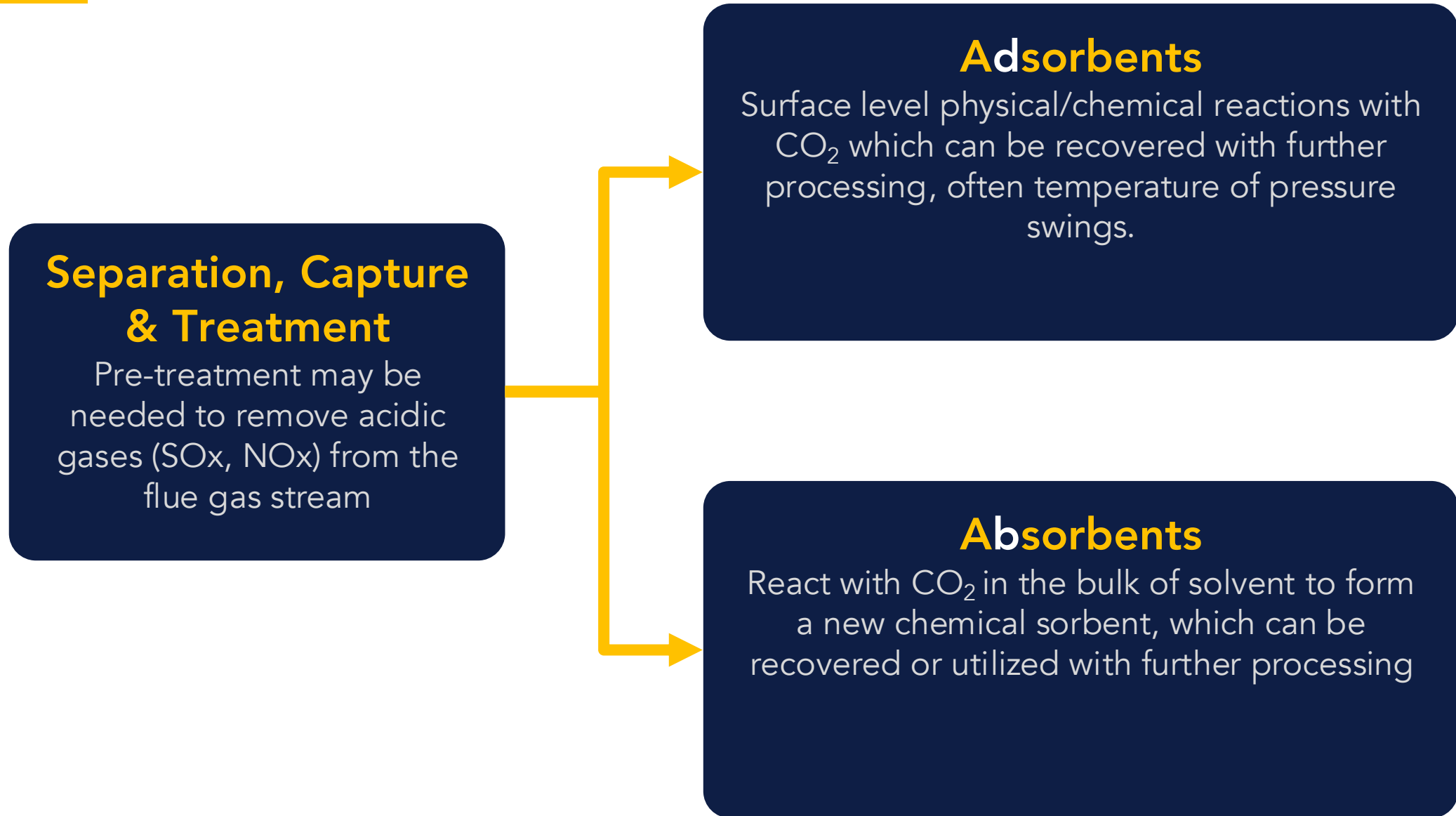


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Selection of Point Source Carbon Capture Technology



Why Rail-Based Carbon Capture?

Rail-Based Carbon Capture

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graph TD; A[Rail-Based Carbon Capture] --> B[Removes Hard-to-Abate Emissions]; A --> C[Lower Energy vs. Direct Air Capture]
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Removes Hard-to-Abate Emissions

- Freight rail emissions are considered "hard-to-abate" due to the difficulties of electrification and long locomotive lifespans
- On-board capture directly addresses diesel exhaust at the source

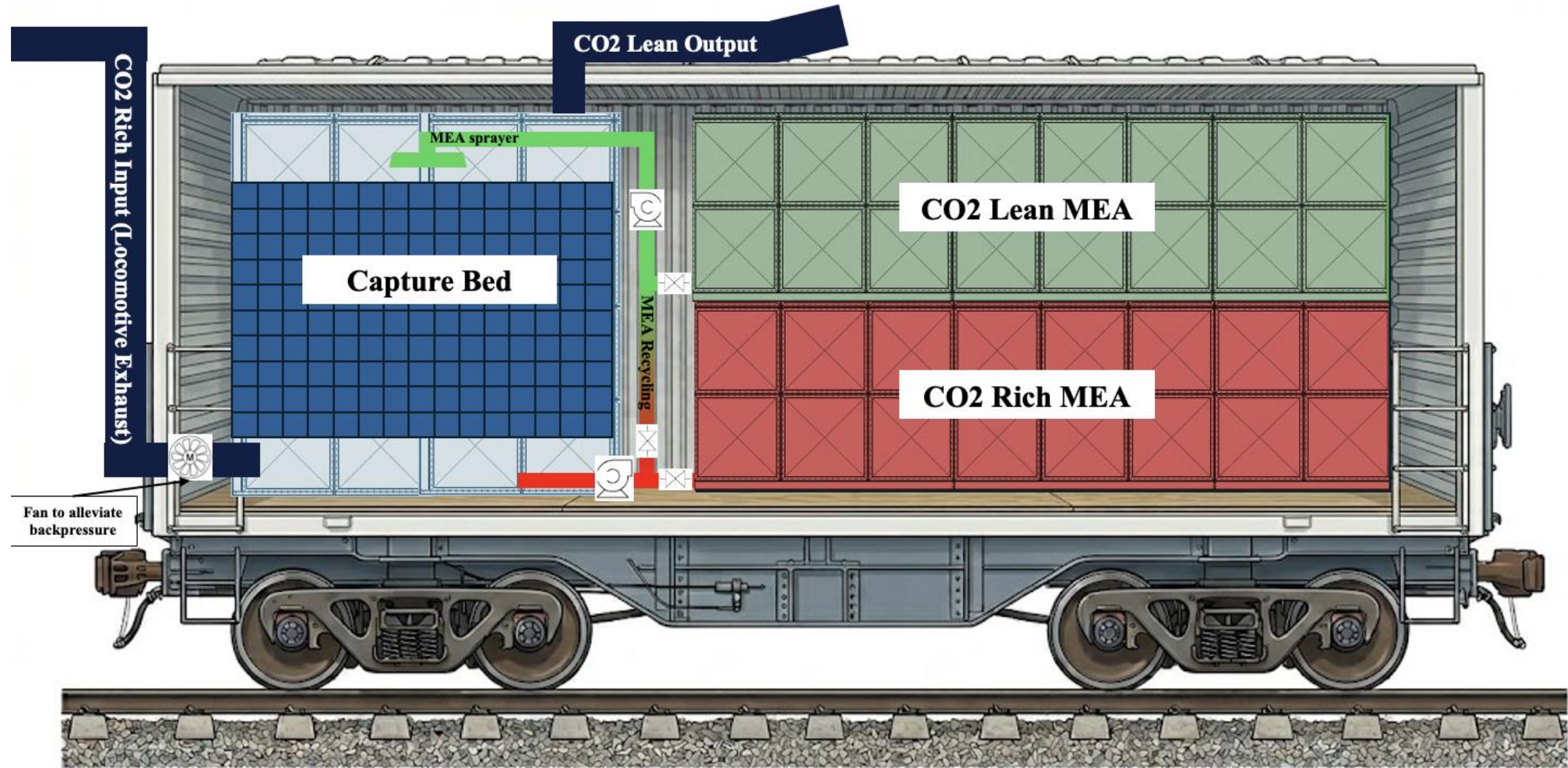
Lower Energy vs. Direct Air Capture

- Concentrated exhaust is easier to capture than ambient air (400 ppm)
- Eliminates the need for large energy-intensive fans required by DAC systems

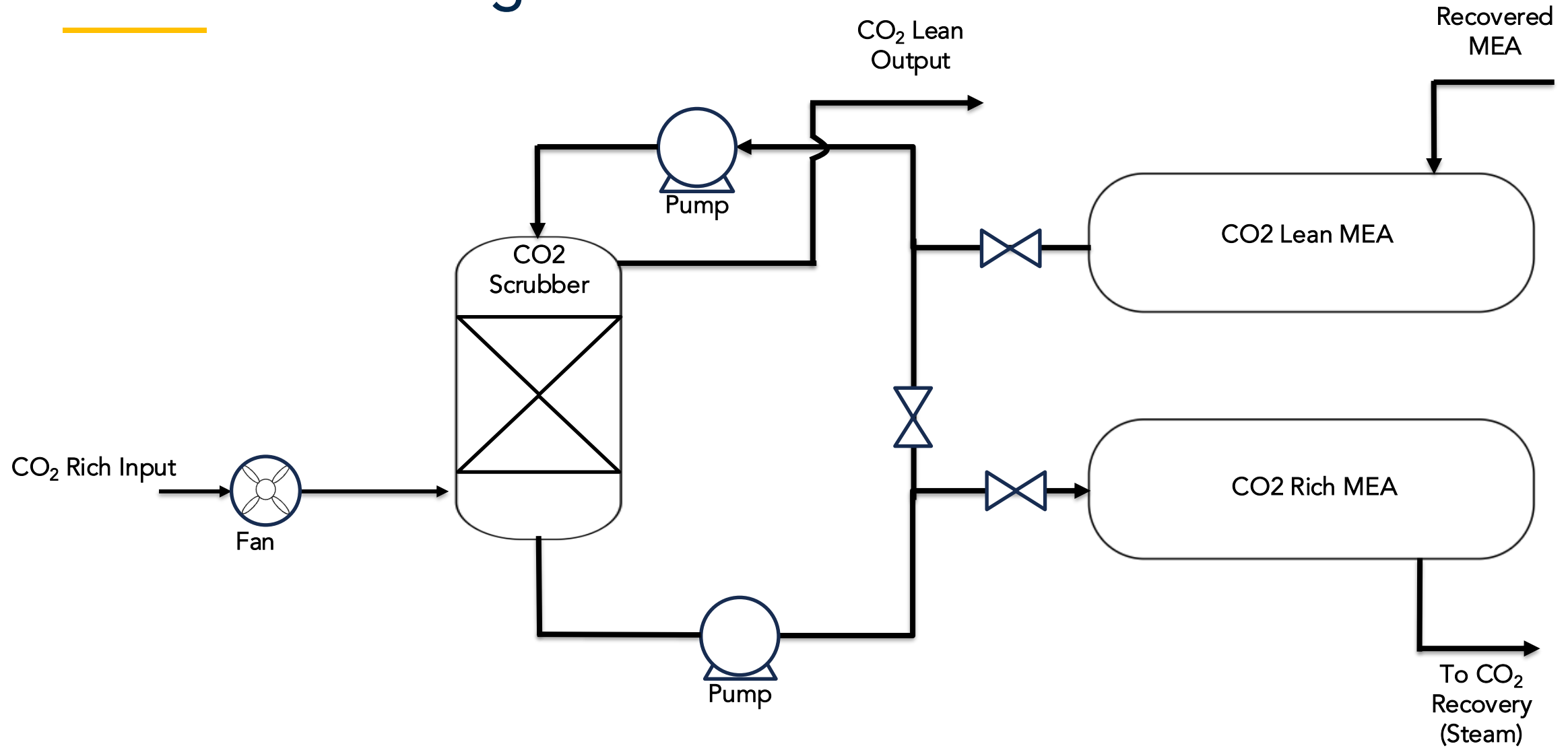
Carbon Capture Technology Selection

	Solvents			Sorbents		
Criteria	Amine Looping (MEA)	Ammonia Looping	Calcium Looping	SAS (Amine)	MOFs	Electroswing
Cost	Low (Amine Looping is widely used for Point Source Capture)	Low (ammonia widely available)	Low (CaO widely available)	~\$1,000/ton	Variable (potentially competitive)	Unknown (Low potential)
CO₂ Selectivity	Very High	High (co-captures)	High but rapidly degrades	Very High	Variable (can be >90%)	High (Expected to perform well at <10% conc. CO ₂)
Lifetime	Mid-tier (MEA degrades)	No degradation if regenerated	Low (rapid decay)	Mid-tier (MEA degrades)	~3% loss per 28 days (Variable)	82% after 215K cycles
Regeneration	100–150°C (thermal)	100–150°C (thermal)	700–900°C (very high)	Vacuum Temperature swing	80–150°C (thermal)	417 kWh/ton (electrical)
Contaminant Resistance	SO ₂ → fertilizer (beneficial)	SO ₂ → fertilizer (beneficial)	SO ₂ → CaSO ₄ (irreversible, co capture)	Sensitive to SO _x , NO _x	Unclear information but varies	O ₂ concern (superoxides)
Vibration Robustness	Moderate Concern (Prone to leakage)	Moderate Concern (Prone to leakage)	Moderate Concern (Prone to leakage)	Good (solid)	Good (solid)	Excellent (solid-state)

Diagram of Proposed Design



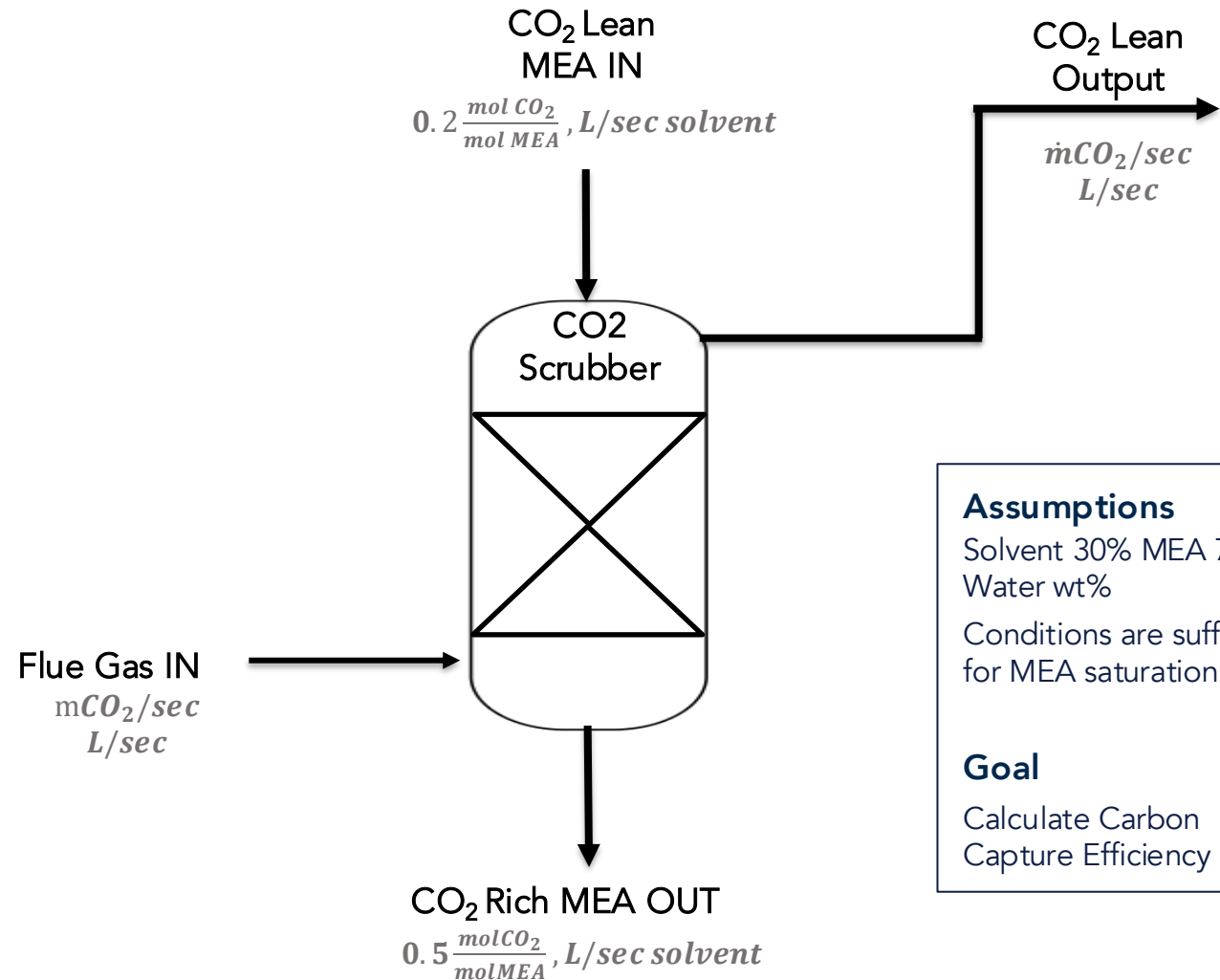
Process Flow Diagram



Life Cycle Assessment

Goal

Provide a rough parametric model for developing a “first pass” LCI for the capture of CO₂ using a liquid solvent on a train. The aim of the LCA more broadly is to determine how viable carbon capture is on a train when compared to alternative decarbonization or defossilization strategies, specifically in this case, either operation of direct air capture off train as part of a “book and claim” type approach or the use of biodiesel.



Assumptions

Solvent 30% MEA 70%
Water wt%
Conditions are sufficient
for MEA saturation

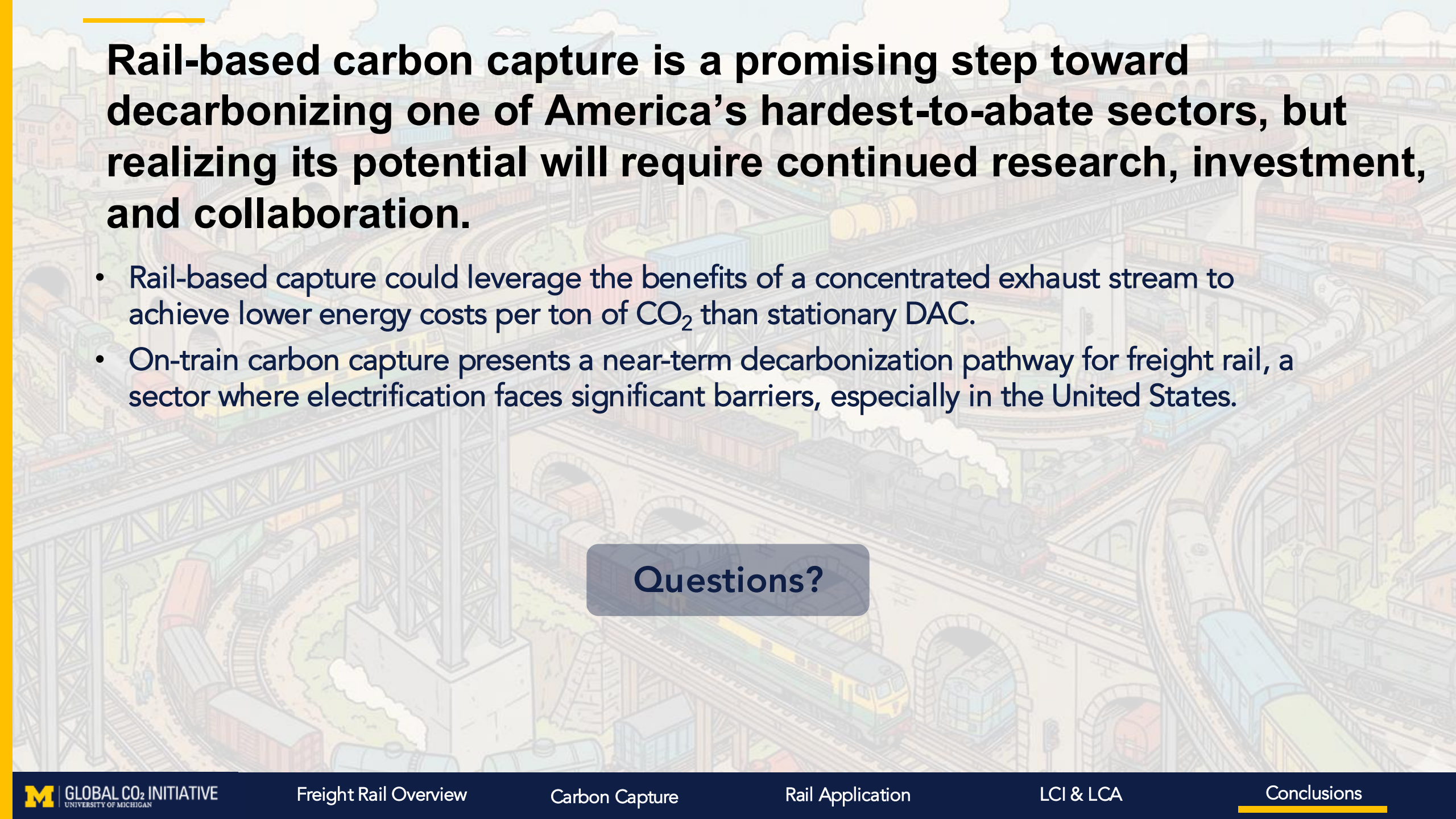
Goal

Calculate Carbon
Capture Efficiency

Preliminary Operational LCI

On-Train Carbon Capture LCI		
OUTPUTS TO TECHNOSPHERE	QTY	Unit
1 Tonne of CO ₂	1,000	kg
INPUTS FROM TECHNOSPHERE	QTY	Unit
MEA Degradation (Market Average Data, Global)	2.2	kg
On-Train Electrical Generation (Diesel Burned in Electric Generating Set)	1.44	MJ
Steam Production for Regeneration (US Midwest Regional Operator)	3,000	MJ
Electricity for Compression (Wind Power, Midwest)	120	kWh
EMISSIONS TO AIR	QTY	Unit
Direct Uncaptured CO ₂ & Other Gases	2422.85	kg

75kg CO₂ emitted per Ton CO₂ Captured



Rail-based carbon capture is a promising step toward decarbonizing one of America's hardest-to-abate sectors, but realizing its potential will require continued research, investment, and collaboration.

- Rail-based capture could leverage the benefits of a concentrated exhaust stream to achieve lower energy costs per ton of CO₂ than stationary DAC.
- On-train carbon capture presents a near-term decarbonization pathway for freight rail, a sector where electrification faces significant barriers, especially in the United States.

Questions?



Appendix

Appendix A: Abstract

The US freight rail network is an important backbone of the economy, moving approximately 1.7 billion tons of goods annually over its expansive network spanning 140,000 miles. It also produces approximately 35 million metric tons of CO₂ a year. Freight rail, especially older switcher trains, emits high levels of harmful pollutants such as SO_x, NO_x, and PM. While rail remains one of the least carbon-intensive forms of freight transportation, rivaled only by maritime shipping, it is particularly difficult to further reduce emissions without significant capital investment. Given the limits of policy, funding, and the privatization of rail, electrification would be incredibly difficult to implement. Locomotives also have long lifespans, with routine remanufacturing extending their service life to over 30 years.

Carbon capture can be a solution to remove emissions, but it can also be an expensive, energy-intensive process. Carbon capture costs tend to be higher for more dilute CO₂ input streams. On-train carbon capture could reduce energy requirements by utilizing its carbon-rich exhaust stream while delivering co-benefits by removing other pollutants. Combining these two technologies to design a rail-based carbon capture system could offer an opportunity to curb hard-to-abate freight rail emissions while also reducing energy requirements, as it requires no turbines.

The purpose of this research is to quantify the environmental impacts associated with building, operating, and disposing of a carbon capture railcar for switcher and long-haul trains. Comparative Life Cycle Assessment (LCA) is used to evaluate the benefits and drawbacks of attaching a carbon capture railcar to a locomotive's exhaust stream versus constructing a separate DAC facility in an area with CO₂ concentrations similar to those of a railyard. Additionally, this research explores the environmental performance of on-train carbon capture compared to other alternatives, like using alternative fuels or electrification.

Keywords: Life Cycle Assessment, Freight Rail, Carbon Capture

Appendix B: Primary Research Questions

1. Can we quantify the environmental impacts associated with rail-based carbon capture?
2. How would a rail-based carbon capture module work? What are its constraints, maintenance requirements, and how would it affect the normal functioning of a train?
3. How would storage and utilization work with train capture? Are there environmental benefits to having carbon capture modules with high mobility (unintended consequences)?
4. From an LCA standpoint, are there alternatives that are more beneficial in terms of carbon capture potential or seeking other decarbonization pathways in the freight-rail industry?

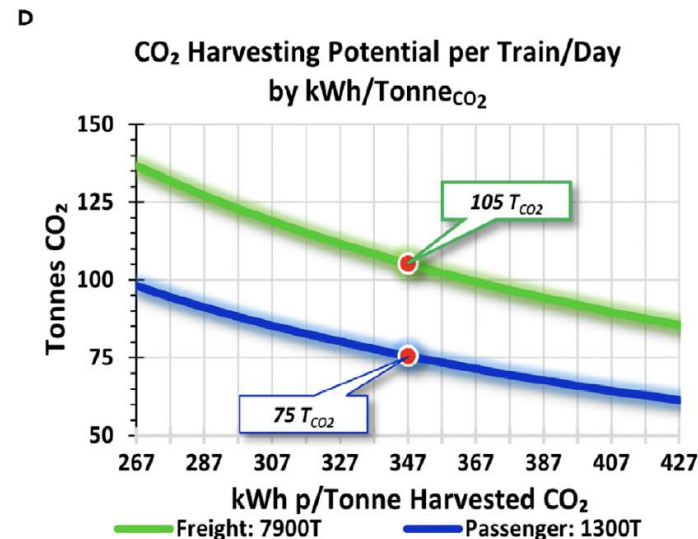
Appendix C: Carbon Capture Concept



Rail-based direct air carbon capture. Bachman, E. et al. Joule, Volume 6, Issue 7, 1368 - 1381

How would a rail-based carbon capture module work? A Proposal

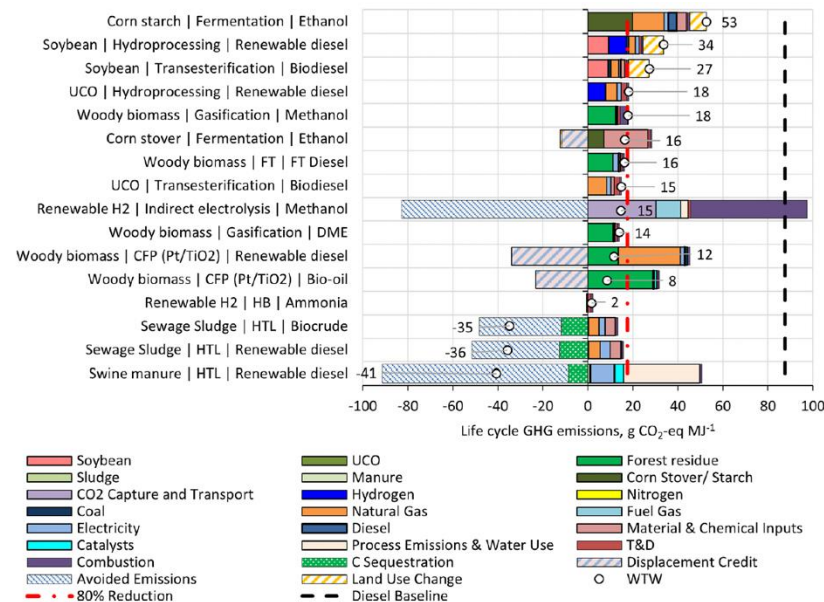
- Capture, Regeneration, onboard storage all done onboard
- Regenerative braking, Solar Arrays + Battery Array are the sources of energy used to regenerate the sorbent technology
- "Will harvest orders of magnitude more CO₂ than is indirectly emitted by the locomotive(s) in additional fuel proximate to their operation from train loading, aerodynamic drag, and rolling resistance."
- "The onboard CO₂ reservoir is emptied into the normal CO₂ tank car located at the station."



Rail-based direct air carbon capture, Bachman, E. et al. Joule, Volume 6, Issue 7, 1368 - 1381

Appendix D: LCA +TEA of Different Fuels to Power Freight Rail

- Fuels that can be used as alternatives can be classified as drop-in, near drop-in & non-drop-in alternatives. Renewable Diesel and Biodiesel would require the least amount of retrofitting to current freight trains (Tier 4 locomotives were studied).
- All alternatives except for biocrude from sludge showed less or comparable SOx emissions and showed similar PM emissions for the most part
- With tax credits, many forms of RD, BD and bio-oils would be considered feasible though their TRLs still between 5&7.



Techno-Economic Analysis and Life Cycle Assessment of Alternative Fuels for Locomotives in the U.S. Freight Rail Sector Tai-Yuan Huang, Farhad H. Masum, Yuan Jiang, Shuyun Li, Scott J. Curran, Munidhar S. Biruduganti, and Troy R. Hawkins *Environmental Science & Technology* 2025 59 (30), 15741-15750 DOI: 10.1021/acs.est.4c11029

LCA +TEA of Different Fuels to Power Freight Rail

Fuel	Feedstock	Conversion	MAC USD\$/MT CO ₂ e	NO _x mg/MJ	PM ₁₀ mg/MJ	CO mg/MJ	TRL
conv. diesel	crude oil	fractional distillation		410	5.6	59	9
biodiesel	soybean	transesterification	80	680	4.8	43	9
biodiesel	UCO	transesterification	120	670	4.2	42	9
RD	soybean	hydroprocessing	150	680	4.8	45	9
RD	UCO	hydroprocessing	100	670	4.0	41	8
RD	sewage sludge	HTL	45	660	8.6	31	5
biocrude	sewage sludge	HTL	−38	640	0.7	11	6
RD	swine manure	HTL	98	670	6.3	42	5
bio-oil	woody biomass	CFP (Pt/TiO ₂)	36	830	2.7	140	7
RD	woody biomass	CFP (Pt/TiO ₂)	61	730	1.4	94	6
RD	woody biomass	Fischer–Tropsch process	120	680	8.9	49	7
methanol	woody biomass	gasification	37	300	2.3	1900	6
methanol	renewable H ₂	indirect electrolysis	490	220	2.5	1900	3
DME	woody biomass	gasification	20	270	2.1	67	5
ethanol	corn starch	fermentation	160	260	8.8	310	9
ethanol	corn stover	fermentation	180	230	8.1	83	7
ammonia	renewable H ₂	Haber–Bosch process	360	190	0.1	310	3

^aMAC (Marginal abatement cost) is calculated from GHG emissions and MFSP for each fuel pathway.

Appendix E: Preliminary Results

Functional Unit CO2	1 Ton of Compressed				
To produce 1 Tonne of CO2					
	Amount	Unit	CO ₂ -eq/unit	Total CO2-eq -	
				kg	
MEA Loss	2.2	kg	3.61	7.942	
On-train electricity usage	1.44	MJ	0.0941	0.135504	
Steam Production For Regeneration	3000	MJ	0.0218	65.4	
Electricity for Compression	120	kWh	0.485	58.2	
Electricity for Cmpression (wind)	120	kWh	0.0134	1.608	
With normal Grid Electricity				133.285504	
With Wind Power				75.085504	
Capture Efficiency from Slide Before	29.22%				
Direct Uncaptured CO2 & Other Gases	707.845981	kg	1	707.8459811	

Appendix E: Preliminary Results

Average MPG Locomotive	0.12 mpg	
Average Speed Freight Train	0.01528 mi/s	
CO2 kg per gallon fuel AVG	8.90 kg/gal	
Fuel Burned Per Second	0.12731 gal/s	
CO2 Mass Flow Rate	1.13 kg/s	
CO2 Mol/sec	25.75 mol/sec	
MEA wt% of Solvent	30.00% %	
Solvent Flow Rate	5.00 L/sec	
Solvent Density	1006.00 kg/m ³	
Mass flow rate Solvent	5.0300 kg/s	
Mass flow rate MEA	1.5090 kg/s	
MEA Mol/s	25.0731 mol/sec	
MEA % Saturated IN	40% %	
MEA % Saturated OUT	100% %	
MEA CO2 mol/ mol MEA IN	0.2	
MEA CO2 mol/ mol MEA OUT	0.5	
OUTPUTS		
CO2 Captured	7.52193274 mol/sec	
Capture Efficiency	29.22% %	

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